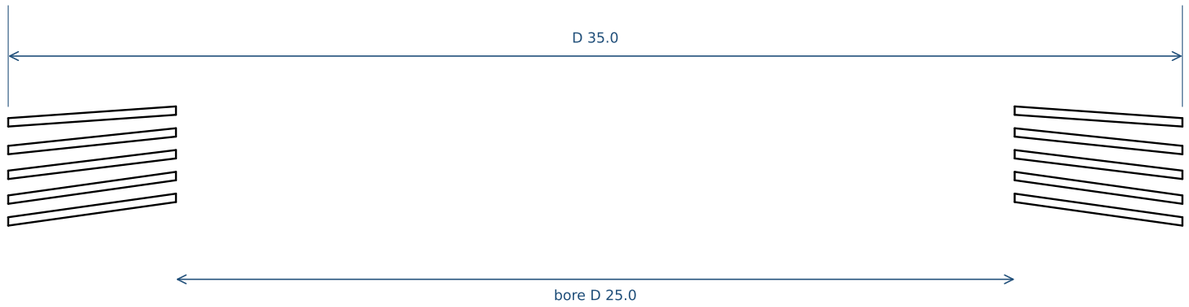
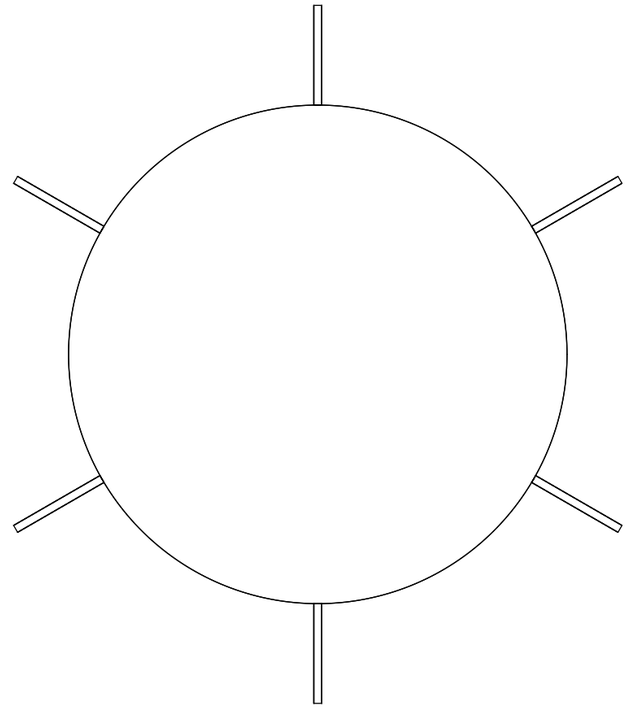
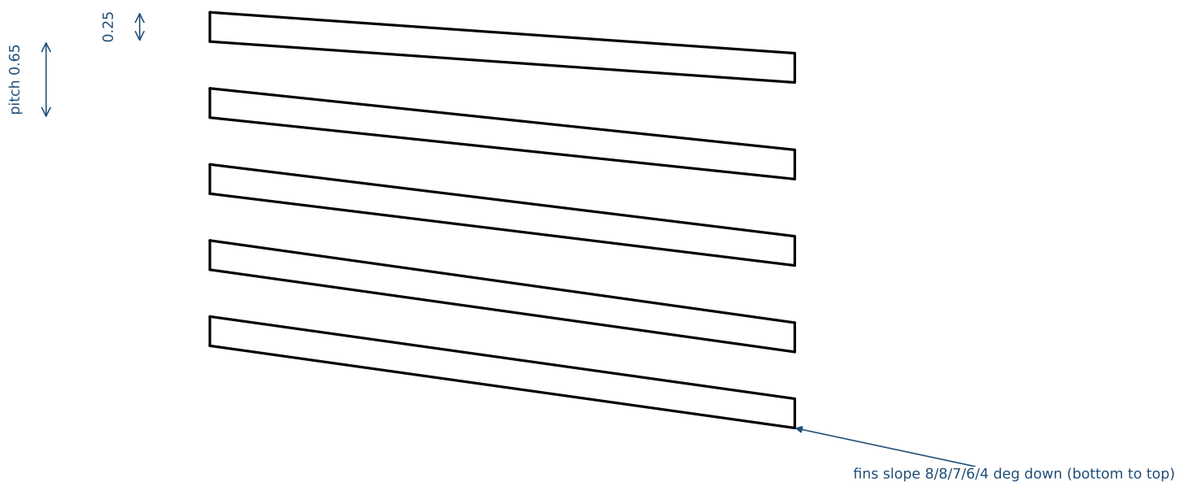
SECTION y=0 (ribs at 30/90/150.. deg are not cut)



TOP VIEW (6 ribs)



DETAIL: fins (x = radius)



NOTES:

1. 5 conical fins 0.25 thick, pitch 0.65, sloping DOWN toward the outside at 8, 8, 7, 6, 4 deg (bottom to top); inner-edge fin bottoms at z 1.90, 2.55, 3.20, 3.85, 4.50 (cap frame). The floor is the core's 8 deg cone. Geometry unchanged from rev A.
2. Channels clear and parallel: open height 0.40 +/-0.05 (check on the first SLA articles; reject sagged fins), fin tilt +/-0.5 deg. This part sets where the light lands.
3. 6 radial ribs 0.4 wide at 30, 90, 150, 210, 270, 330 deg join the fins; rib bottoms follow the 8 deg floor.
4. Top face at z 4.75 +0/-0.05. The dome lip sits 0.25 ABOVE it (rev A 0.05): the louver is never clamped. Retained by 3 dots of UV adhesive on the floor.
5. No gloss: fin surfaces matte (Ra > 1.6 um); a glossy fin re-reflects light at the wrong angle.

RECORD SCONCE | LABEL-LIGHT LOUVER RING (angular filter)

PART NO: RS-CAP-12L-LOUVER REV: B DATE: 2026-11-11 SUPERSEDES: rev A, A.1 (do not build or quote)

MATERIAL: Prototype: precision SLA in black resin (BMF microArch / Formlabs Form 4 Black, 25 um). Production: micro-moulded PC, black.

FINISH: Fins matte black. Inner 30% of every fin (r 12.5..14.0) white: dip the bore edge in white reflector ink 1.5 mm deep, or 2-shot mould.

UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown (enlarged detail at right)

GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted